

Topics: 12.1 Three-Dimensional Coordinate Systems; 12.2 Vectors; 12.3 The Dot Product

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1. Consider the point $P(3, 4, 5)$.

- (a) What is the projection of the point onto the xy -plane?

The projection is $(3, 4, 0)$.

- (b) What is the projection of the point onto the xz -plane?

The projection is $(3, 0, 5)$.

- (c) Find the length of the line segment $\overline{OP}$, where $O(0, 0, 0)$ is the origin.

The length is the distance between the origin $O(0, 0, 0)$ and $P(3, 4, 5)$, given by the following.

$$\sqrt{(3-0)^2 + (4-0)^2 + (5-0)^2} = \sqrt{50} = 5\sqrt{2}.$$

- (d) Find the position vector for the point $P(3, 4, 5)$ in **ijk**-form.

The position vector has initial point at the origin O and terminal point at P . So, the vector $\overrightarrow{OP}$ from the origin $O(0, 0, 0)$ to the point $P(3, 4, 5)$ is

$$(3-0)\mathbf{i} + (4-0)\mathbf{j} + (5-0)\mathbf{k} = 3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}.$$

2. Find an equation of a sphere if one of its diameters has endpoints at $(1, 2, 4)$ and $(4, 3, 10)$.

First, we find the sphere's diameter by computing the distance between the two given points, $(1, 2, 4)$ and $(4, 3, 10)$. We have

$$d = \sqrt{(1-4)^2 + (2-3)^2 + (4-10)^2} = \sqrt{46}$$

Since the radius, r , of the sphere, is half the diameter, we obtain $r = \frac{d}{2} = \frac{\sqrt{46}}{2}$.

Now the center of the sphere, C , is located at the midpoint of its diameter. To find the coordinates of the midpoint of a segment, we take the average of the corresponding coordinates of the end points of the segment:

$$\left(\frac{4+1}{2}, \frac{2+3}{2}, \frac{4+10}{2} \right).$$

Hence, the center is located at

$$C \left(\frac{5}{2}, \frac{5}{2}, 7 \right).$$

Thus our sphere has radius $r = \frac{\sqrt{46}}{2}$ with a center at $\left(\frac{5}{2}, \frac{5}{2}, 7\right)$. Therefore its equation can be written as

$$\left(x - \frac{5}{2}\right)^2 + \left(y - \frac{5}{2}\right)^2 + (z - 7)^2 = \frac{23}{2}.$$

3. Given $\mathbf{a} = 9\mathbf{i} - 8\mathbf{j} + 7\mathbf{k}$ and $\mathbf{b} = \langle 7, 0, -9 \rangle$, find the following. Simplify your answers completely.

(a) $\mathbf{a} + \mathbf{b}$

$$\mathbf{a} + \mathbf{b} = \langle 9, -8, 7 \rangle + \langle 7, 0, -9 \rangle = \langle 16, -8, -2 \rangle.$$

(b) $3\mathbf{a} - \mathbf{b}$

$$3\mathbf{a} - \mathbf{b} = 3\langle 9, -8, 7 \rangle - \langle 7, 0, -9 \rangle = \langle 27, -24, 21 \rangle - \langle 7, 0, -9 \rangle = \langle 20, -24, 30 \rangle.$$

(c) $|\mathbf{b}|$

$$|\mathbf{b}| = \sqrt{7^2 + 0^2 + (-9)^2} = \sqrt{49 + 81} = \sqrt{130}.$$

(d) $|\mathbf{b} - \mathbf{a}|$

$$|\mathbf{b} - \mathbf{a}| = | \langle -2, 8, -16 \rangle | = \sqrt{(-2)^2 + 8^2 + (-16)^2} = \sqrt{4 + 64 + 256} = \sqrt{324} = 18.$$

4. Find the vector that has the opposite direction to $\langle 9, -6, -2 \rangle$ and has length 5.

Since

$$|\langle 9, -6, -2 \rangle| = \sqrt{9^2 + (-6)^2 + (-2)^2} = \sqrt{121} = 11,$$

the unit vector in the opposite direction of $\langle 9, -6, -2 \rangle$ is

$$\mathbf{u} = -\frac{1}{11}\langle 9, -6, -2 \rangle = \left\langle -\frac{9}{11}, \frac{6}{11}, \frac{2}{11} \right\rangle.$$

The vector in the opposite direction and with length 5 is

$$5\mathbf{u} = 5\left\langle -\frac{9}{11}, \frac{6}{11}, \frac{2}{11} \right\rangle = \left\langle -\frac{45}{11}, \frac{30}{11}, \frac{10}{11} \right\rangle.$$

5. Determine whether the given vectors are orthogonal, parallel, or neither.

(a) $\mathbf{a} = 4\mathbf{i} - \mathbf{j} + 4\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i} + 12\mathbf{j} - 2\mathbf{k}$

Orthogonal: We have

$$\mathbf{a} \cdot \mathbf{b} = 4 \cdot 5 + (-1) \cdot 12 + 4 \cdot (-2) = 0,$$

so $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

(b) $\mathbf{a} = \langle 6, 5, -2 \rangle$ and $\mathbf{b} = \langle 5, 0, 9 \rangle$

Neither: We have

$$\mathbf{a} \cdot \mathbf{b} = 6 \cdot 5 + 5 \cdot 0 + (-2) \cdot 9 \neq 0,$$

so the vectors are not orthogonal. Since $\mathbf{a}$ is not a scalar multiple of $\mathbf{b}$, they are not parallel.

(c) $\mathbf{a} = \langle -18, 15 \rangle$ and $\mathbf{b} = \langle 12, -10 \rangle$

Parallel: We see that $\mathbf{b} = -\frac{2}{3}\mathbf{a}$, so the vectors are parallel (with opposite direction).

6. Given vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ in V_3 , which of the following expressions are meaningful? Which are meaningless? Explain your reasoning.

(a) $(\mathbf{a} \cdot \mathbf{b}) \cdot \mathbf{c}$

The expression $\mathbf{a} \cdot \mathbf{b}$ is a scalar and $\mathbf{c}$ is a vector. One cannot take the dot product of a scalar and a vector, so the expression is meaningless.

(b) $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c}$

Both $\mathbf{a} + \mathbf{b}$ and $\mathbf{c}$ are vectors, so the dot product $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c}$ has meaning.

(c) $\mathbf{a} + (\mathbf{b} \cdot \mathbf{c})$

The expression $\mathbf{b} \cdot \mathbf{c}$ is a scalar, while $\mathbf{a}$ is a vector. A vector and a scalar cannot be added, so the expression is meaningless.

7. Find the angle θ between the vectors $\mathbf{a} = \langle 2, 0, -3 \rangle$ and $\mathbf{b} = -2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$.

We first compute the dot product:

$$\mathbf{a} \cdot \mathbf{b} = \langle 2, 0, -3 \rangle \cdot \langle -2, 3, 1 \rangle = 2(-2) + 0(3) + (-3)1 = -7.$$

Therefore

$$\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} = \frac{-7}{\sqrt{2^2 + 0^2 + (-3)^2} \sqrt{(-2)^2 + 3^2 + 1^2}} = \frac{-7}{\sqrt{13} \sqrt{14}}.$$

So, the answer is $\theta = \arccos\left(\frac{-7}{\sqrt{13} \sqrt{14}}\right)$.

8. Consider $\mathbf{a} = \langle -1, 4, 8 \rangle$ and $\mathbf{b} = \langle 18, 2, 1 \rangle$.

- (a) Find the scalar projection of $\mathbf{b}$ onto $\mathbf{a}$.

Since $|\mathbf{a}| = \sqrt{1 + 16 + 64} = \sqrt{81} = 9$, the scalar projection of $\mathbf{b}$ onto $\mathbf{a}$ is

$$\text{comp}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|} = \frac{(-1) \cdot 18 + 4 \cdot (2) + 8 \cdot (1)}{9} = -\frac{2}{9}.$$

- (b) Find the vector projection of $\mathbf{b}$ onto $\mathbf{a}$.

The vector projection of $\mathbf{b}$ onto $\mathbf{a}$ is

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \text{comp}_{\mathbf{a}} \mathbf{b} \frac{\mathbf{a}}{|\mathbf{a}|} = -\frac{2}{9} \frac{\mathbf{a}}{|\mathbf{a}|} = -\frac{2}{9} \frac{\langle -1, 4, 8 \rangle}{9} = \left\langle \frac{2}{81}, -\frac{8}{81}, -\frac{16}{81} \right\rangle$$

9. Find the work (in joules) done by the constant force $\mathbf{F} = 8\mathbf{i} - 6\mathbf{j} + 5\mathbf{k}$ acting on an object that moves from the point $(0, 6, 4)$ to the point $(4, 14, 22)$ along a straight line. The distance is measured in meters and the force in newtons.

The displacement vector is

$$\mathbf{D} = (4 - 0)\mathbf{i} + (14 - 6)\mathbf{j} + (22 - 4)\mathbf{k} = 4\mathbf{i} + 8\mathbf{j} + 18\mathbf{k}.$$

Since $W = \mathbf{F} \cdot \mathbf{D}$, the work done is

$$W = \mathbf{F} \cdot \mathbf{D} = 8 \cdot 4 + (-6) \cdot 8 + 5 \cdot 18 = 32 - 48 + 90 = 74 \text{ joules.}$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Dot Product If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

If θ is the angle between nonzero vectors $\mathbf{a}$ and $\mathbf{b}$, then

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta.$$

Scalar and Vector Projections If $\mathbf{a} \neq \mathbf{0}$, then the *scalar projection* of $\mathbf{b}$ onto $\mathbf{a}$ is

$$\text{comp}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|}.$$

The *vector projection* of $\mathbf{b}$ onto $\mathbf{a}$ is

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|} \right) \frac{\mathbf{a}}{|\mathbf{a}|} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|^2} \right) \mathbf{a}.$$

Work by a Constant Force If a constant force $\mathbf{F}$ moves an object through displacement $\mathbf{D}$, then the work done is

$$W = \mathbf{F} \cdot \mathbf{D}.$$

Suggested Textbook Problems

Section 12.1: 1-46

Section 12.2: 1-33, 37-44

Section 12.3: 1-56